MMA Decision Lab

Research record · Exploratory baseline

Focal fighter · Fighter A

Khabib Nurmagomedov

vs

Opponent

Conor McGregor

Before **2018-10-06** · Lightweight · 5 scheduled rounds
Snapshot greco-5e38ebeb9bb8 · Model 0.1.0 · Analysis created 2026-10-16T12:23:48-05:00

Analysis identity: be8c0f6ae9baa193f83be655c547366000e8f3b6d5048ccd8ca163745130f200

Exploratory baseline

No robust tactical preference under the tested assumptions.

Striking priority has the highest modeled mean utility. A preference requires supported baseline evidence, a positive paired 95% advantage above the declared margin, and stability throughout the tested grid. Tactical effects remain assumptions; this is not causal evidence of an optimal real-world strategy.

Baseline source: Historical research. Historical forecasts and hypothetical tactical effects have different evidential status.

Decision and applied assumptions

The focal fighter and coaching team examine static pre-fight priorities. Within this binary model U(win)=1 and U(loss)=0, so expected utility equals modeled win probability. These allocations do not guarantee execution or time in a fight state.

Assumed static allocations; planned opportunities, not observed fight time

Tactical priority	Planned opportunity emphasis	Standing	Clinch	Ground
Striking priority	Distance management and standing exchanges; limited intentional grappling.	75%	15%	10%
Grappling priority	Takedown entries, clinch pressure and ground offense.	20%	30%	50%

Tactical priority	Planned opportunity emphasis	Standing	Clinch	Ground
Balanced priority	Static combination of standing, clinch and ground emphasis.	45%	25%	30%

Reference allocation: Standing 50% / Clinch 25% / Ground 25%. This engineering origin is separate from Balanced priority and from observed historical profiles.

Applied tactical and uncertainty assumptions

Assumed parameter	Applied value	Unit
Assumed striking effect	0.35	log-odds
Assumed grappling effect	0.5	log-odds
Opponent resistance index	0.4	dimensionless
Counter-exposure cost	0.15	log-odds
Lower / expected / stronger response weights	0.25 / 0.5 / 0.25	probabilities
Triangular effect half-width	0.3	log-odds
Dirichlet response concentration	30	dimensionless

All applied effects, resistance, response weights and uncertainty settings are assumed (Category C), not fitted interventions. Lower / expected / stronger resistance offsets are −0.25 / 0 / +0.25, clipped to [0,1]. Response weights are exogenous to policy. Identically neutral effects require s=g=c=0 and effect half-width=0.

Earlier-date evidence and baseline

68.584% Estimated P(Fighter A wins)

Conditional bootstrap parameter interval: 62.898%–75.500%; 64 fitted bootstrap models. This is a conditional parameter approximation, not a complete predictive confidence interval.

Exploratory baseline: held-out evidence is insufficient against the 0.5 reference.

Earlier-date fighter evidence; common metrics and units

Group	Metric	Khabib Nurmagomedov	Conor McGregor
Experience	Earlier eligible bouts	10	10
Experience	Observed exposure (min)	135.20	81.30
Experience	Last earlier appearance	2018-04-07	2016-11-12
Experience	Last earlier division	Lightweight	Lightweight
Experience	Inactivity (years)	0.50	1.90
Striking	Significant strikes landed / min	4.34	5.82

Group	Metric	Khabib Nurmagomedov	Conor McGregor
Striking	Significant strikes absorbed / min	1.55	4.55
Striking	Smoothed striking accuracy	48.8%	47.6%
Striking	Smoothed striking defense	70.3%	57.3%
Wrestling	Takedowns landed / 15 min	5.44	0.92
Wrestling	Takedown attempts / 15 min	12.09	1.48
Wrestling	Smoothed takedown accuracy	45.0%	60.0%
Wrestling	Smoothed takedown defense	80.0%	72.0%
Grappling	Submission attempts / 15 min	0.55	0.00
Grappling	Control seconds / observed min	32.26	9.24

Only observations strictly before 2018-10-06 contribute. Rates are estimated from retained history; displayed ratios use the documented smoothing. Missing observations remain missing. Descriptive differences do not establish tactical advantages; a higher absorbed-strike rate is not automatically favorable.

Forecasting evidence

Untouched chronological test; accuracy is a secondary measure

Model	Brier score ↓	Log loss ↓	Accuracy	AUC
Logistic baseline	0.24252	0.67808	57.1%	0.60
50/50 reference	0.25000	0.69315	50.1%	0.50
Win-rate-only reference	0.24400	0.68107	54.3%	0.59

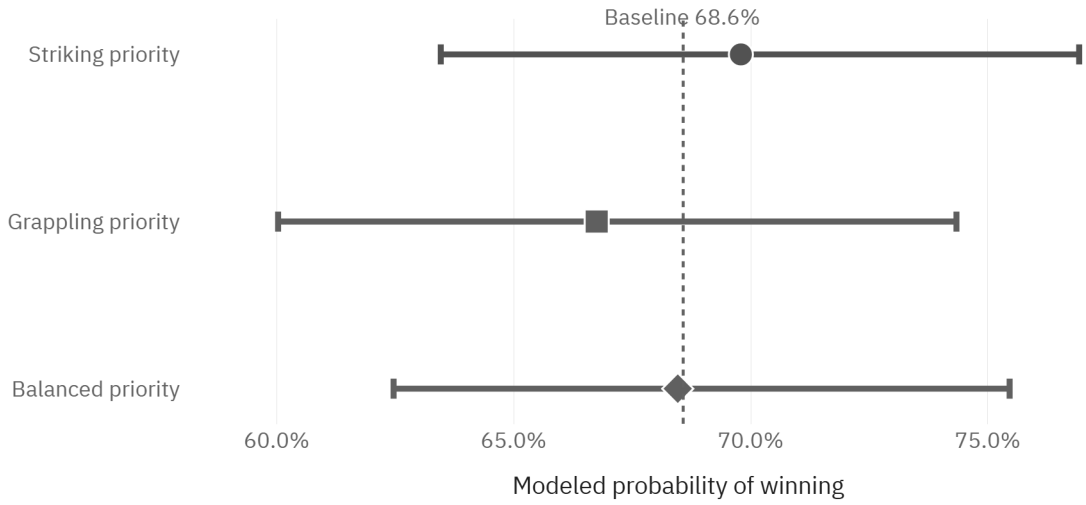
Chronological evaluation partitions

Partition	Bouts	First date	Last date
test	527	2016-11-19	2018-09-22
train	1257	2000-09-22	2015-01-31
validation	493	2015-02-14	2016-11-12

95% iid bout-bootstrap interval for Brier(model) − Brier(50/50): **-0.01620 to +0.00077**. Calibration remains descriptive; repeated-fighter dependence is not captured.

Modeled outcomes and uncertainty

Expected utility · central 95% bands



Calculated policy outcomes; pp = percentage points

Tactical priority	Mean modeled P(win)	Central 95% band	Change from paired baseline (pp)	Rank-first share	Expected regret (pp)	MC standard error (pp)
Striking priority	69.792%	63.462%–76.928%	+0.998	94.9%	0.015	0.0502
Grappling priority	66.751%	60.031%–74.339%	-2.043	0.4%	3.056	0.0526
Balanced priority	68.461%	62.466%–75.461%	-0.333	4.7%	1.346	0.0502

Central 95% bands describe the specified parameter and assumption distribution, not uncertainty in a causal effect or the frequency of future win/loss observations. The dotted baseline is the central reference estimate; change from baseline uses paired baseline draws.

Controlled uncertainty comparisons; these SDs are not additive variance components

Tactical priority	Empirical-only SD (pp)	Scenario-only SD (pp)
Striking priority	3.464	0.677
Grappling priority	3.649	0.869
Balanced priority	3.550	0.141

Paired comparisons

Paired utility differences from the same integration draws

Priority A	Priority B	Mean A – B (pp)	2.5% quantile (pp)	97.5% quantile (pp)	MC standard error (pp)

Priority A	Priority B	Mean A – B (pp)	2.5% quantile (pp)	97.5% quantile (pp)	MC standard error (pp)
Striking priority	Grappling priority	+3.0409	+0.1157	+6.0287	+0.0217
Striking priority	Balanced priority	+1.3311	-0.2277	+2.8989	+0.0115
Grappling priority	Balanced priority	-1.7098	-3.1340	-0.3241	+0.0102

Rank-first shares are conditional on this uncertainty model; exact ties share mass equally. Expected regret compares each policy with the best within the same common parameter draw. Future fight outcome randomness is integrated analytically.

5,000 integration draws · Seed 422. Draw count is computational; increasing it reduces numerical error and adds no historical evidence.

Decision-tree calculation

Decision: tactical priority → Chance: opponent response → Outcome: win/loss → Utility: 1/0. Expected utility is the sum of the calculated terminal contributions below; the tree uses central assumptions.

Calculated decision → response → outcome → utility branches

Priority	Opponent response	Response probability	Conditional outcome probability	Outcome	Utility	Joint probability	Utility contribution
Striking priority	Lower resistance	0.250000	0.684623	Win	1	0.171156	0.171156
Striking priority	Lower resistance	0.250000	0.315377	Loss	0	0.078844	0.000000
Striking priority	Expected resistance	0.500000	0.695980	Win	1	0.347990	0.347990
Striking priority	Expected resistance	0.500000	0.304020	Loss	0	0.152010	0.000000
Striking priority	Stronger resistance	0.250000	0.707102	Win	1	0.176776	0.176776
Striking priority	Stronger resistance	0.250000	0.292898	Loss	0	0.073224	0.000000
Grappling priority	Lower resistance	0.250000	0.684381	Win	1	0.171095	0.171095
Grappling priority	Lower resistance	0.250000	0.315619	Loss	0	0.078905	0.000000
Grappling priority	Expected resistance	0.500000	0.665461	Win	1	0.332731	0.332731
Grappling priority	Expected resistance	0.500000	0.334539	Loss	0	0.167269	0.000000

Priority	Opponent response	Response probability	Conditional outcome probability	Outcome	Utility	Joint probability	Utility contribution
Grappling priority	Stronger resistance	0.250000	0.645994	Win	1	0.161499	0.161499
Grappling priority	Stronger resistance	0.250000	0.354006	Loss	0	0.088501	0.000000
Balanced priority	Lower resistance	0.250000	0.685594	Win	1	0.171399	0.171399
Balanced priority	Lower resistance	0.250000	0.314406	Loss	0	0.078601	0.000000
Balanced priority	Expected resistance	0.500000	0.682487	Win	1	0.341244	0.341244
Balanced priority	Expected resistance	0.500000	0.317513	Loss	0	0.158756	0.000000
Balanced priority	Stronger resistance	0.250000	0.679364	Win	1	0.169841	0.169841
Balanced priority	Stronger resistance	0.250000	0.320636	Loss	0	0.080159	0.000000

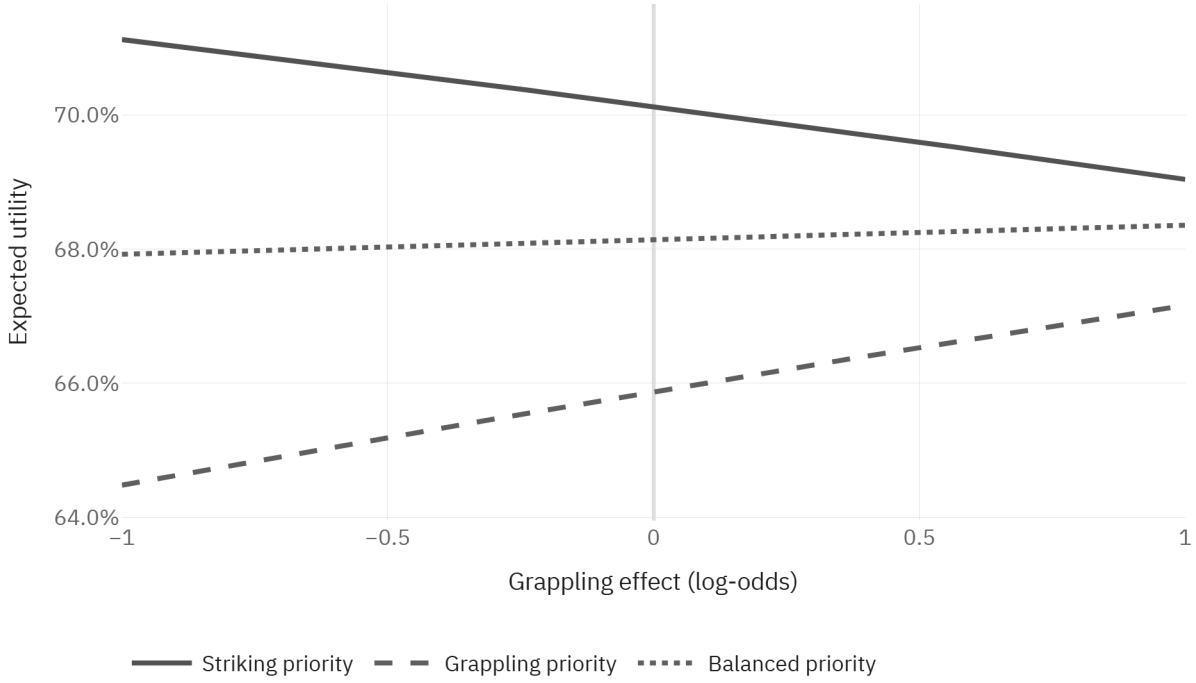
Default sensitivity and robustness

This record includes the engine's default grappling-effect one-way sweep and grappling-effect × resistance decision grid. Sensitivity grids explored separately in the interface are separate records and are not substituted into this report or its robustness criterion.

Declared default stress ranges in this record

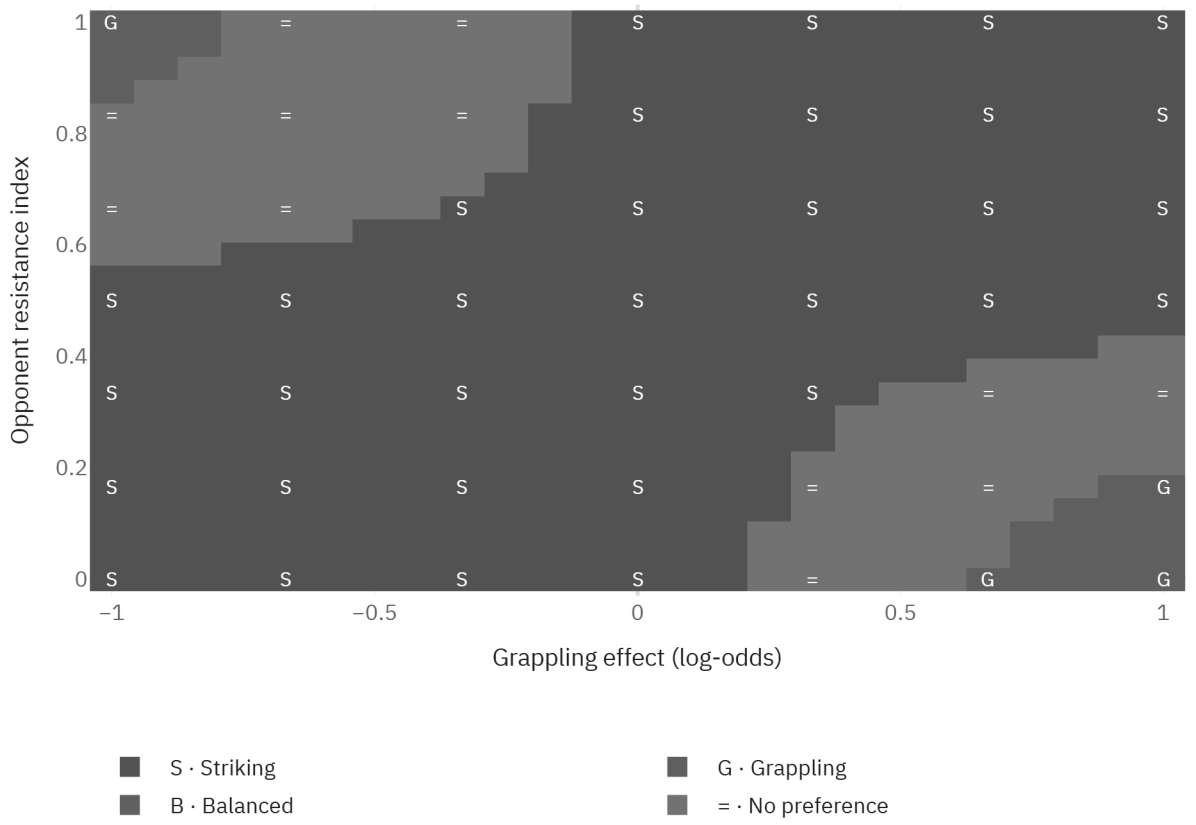
Parameter	Minimum	Maximum
Assumed striking effect	-1	1
Assumed grappling effect	-1	1
Opponent resistance index	0	1
Counter-exposure cost	0	0.5

Sensitivity to one assumption



No unique top-ranking switching threshold was found within the recorded one-way range.

Decision regions · tested assumptions



Pointwise regions require a leading central utility gap above 0.01. S = Striking, G = Grappling, B = Balanced; the code = means no pointwise preference. Pointwise preference is distinct from full uncertainty robustness.

A conditional preferred strategy additionally requires supported held-out baseline evidence, a paired 2.5% advantage against every competitor above the declared 0.01 margin plus its numerical-error allowance, and the same leader throughout the default tested grid. These are engineering criteria, not universal scientific thresholds.

Recorded robustness criteria and paired diagnostics

Recorded criterion or diagnostic	Value
Held-out baseline support criterion met	No
Lowest paired 2.5% advantage (pp)	-0.2277
Maximum paired MC standard error (pp)	0.0217
Same leader throughout default tested grid	No
Declared utility margin (pp)	1.00

Limitations and interpretation

- All policy allocations are engineering presets; execution feasibility has not been established by coaching evidence.
- Tactical effects, allocations, opponent responses and uncertainty distributions are assumptions; no causal tactic effect is identified. Balanced is static, not a learned adaptive policy.
- This implementation uses a logistic baseline, not the full state-transition Markov model of Holmes et al. The reference research already uses simulation.
- Current archival records may contain later corrections; strict event dates do not establish an as-was publication-time backtest.
- Retained earlier bouts are not complete professional career records. Nonbinary, DQ and unsupported-format bouts are excluded. Full primary-source reconciliation and reuse rights remain outstanding.
- The baseline pools divisions and 3/5-round formats and does not estimate individualized division, bout-length, opponent-strength, injury, preparation or changing-form effects.
- Bout bootstrap ignores repeated-fighter dependence and conditions on the selected regularization and fixed historical features. Structural and empirical draws are assumed independent; the displayed components do not add to total variance.
- Pointwise sensitivity regions use central assumptions. Robustness is limited to the declared paired criterion and finite default grid, not all possible tactics, models or parameter ranges.

Data quality and provenance

Observed snapshot coverage and ingestion counts

Snapshot measure	Value
Source result rows	8,962
Retained eligible bouts	8,420
Excluded source rows	542
Competitor-round records	40,068
Source fighter identities	4,628
First retained date	1999-07-16
Last retained date	2026-10-10

Source: UFCStats via community CSV archive
Reuse / license status: Repository declares GPL-3.0; original-source rights not independently established. Local research copy; no data redistribution in Git.

Frozen input file hashes

Source file	SHA-256
ufc_event_details.csv	f8dc81682ed176b1a6bb32dfee19cdbe41eca704db57566d60f74ae43052cc40
ufc_fight_results.csv	e1616c4378d78771d4e308df67a9f1dcc41fa951a21783d1864eef2e94156974
ufc_fight_stats.csv	758ca57cb4c11b9cfdd33c649c8938ddff920495339e5070014590eb17865c2f
ufc_fighter_details.csv	a1f6e4326217fab9734ac42f1958cd07b0b2b98f3caa5da08497ec7648eb6f70
LICENSE	3972dc9744f6499f0f9b2dbf76696f2ae7ad8af9b23dde66d6af86c9dfb36986
README.md	c06ba3cc5d4586ebab9485f08b8f444d8171c364281fc01a97a63302405e979f

Analytical implementation SHA-256:
72b0d5a109086f954a65e562906ede7ff3f7c09ce3576f893d2b5128320e9305

Research foundations

- Holmes, McHale & Żychaluk (2023). A Markov chain model for forecasting results of mixed martial arts contests.
- Miarka, Brito & Amtmann (2017). Performance probabilities and outcome effects in technical–tactical factors with bout phase changes of mixed martial arts.

- James et al. (2017). Identifying the performance characteristics of a winning outcome in elite mixed martial arts competition.
- Bueno et al. (2022). Exploratory systematic review of mixed martial arts.
- Gneiting & Raftery (2007). Strictly proper scoring rules, prediction, and estimation.
- Greco1899. Published UFCStats community CSV archive (GPL-3.0 declared).

References provide context; they do not imply endorsement or validate this implementation.

Independent academic research prototype. Not affiliated with or endorsed by UFC. Educational decision analysis; no wagering functionality. This HTML embeds its fonts and plotting runtime for offline reading. Structured JSON and CSV retain the numerical analysis record.